What Supervision Is Identifiable in Single-Image Reflection Removal? A Tinted-Glass Benchmark and a Diagnostic Study

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***Abstract*—Real-pair supervised single-image reflection removal (SIRR) trains on images whose reflection-free target is captured under a protocol, and every protocol silently assumes something about the reflection layer. We examine those assumptions for *automotive tinted glass*, where a camera-side occluder (the photographer) cuts part of the reflected solid angle *and* part of the illumination. Measuring 82 human-confirmed pairs, the two effects are *not* the same size and do not even point the same way for the two layers: the reflection-free target is only 3.7% darker inside Ω, while the reflected content is 16.8% weaker, so the input is 13% darker than its own target there ($I/T|_{\text{in}} \approx 0.87$). A correction applied at one global strength therefore fails inside the occluded region; the resulting over-correction ratio is measured at $2.7\times$. Because the occlusion region cannot be localised from a single frame (IoU ceiling 0.494 in-domain, 0.14 across domains), we do not score methods inside it on the test panel at all**

We contribute (i) a tinted-glass benchmark of 631 pairs with four controlled attenuation levels, per-frame occlusion masks and a signed reflection layer, plus zero-shot baselines for seven public methods; (ii) a formalization of the occlusion region and a proof that the global photometric scale is not identifiable from a single frame; (iii) a diagnostic of where supervision should be placed: with three seeds, an area-matched *random* region (19.90 dB) matches the true occlusion mask (19.41 dB, difference -0.49 dB, not significant), and a whole-image relative error already reaches 20.05 dB, so the active ingredient is the reflection-normalized error rather than the region identity, while a closed-form support-area bound (0.036 dB) shows why boundary-scale terms cannot be validated at all; and (iv) a transfer study: fine-tuning on our data gains $+2.22$ dB over the best public method on 15/15 scenes and still $+1.69$ dB after a per-image affine, and the gain is largest for the pool that contains *the fewest* occlusion regions — what transfers is the domain, not the occlusion. An interventional protocol shows that state-of-the-art photometric conditioning fails its own requirement (0.02 vs. 0.50).

***Index Terms*—single-image reflection removal, tinted glass, occlusion region, identifiability, supervision placement, benchmark**

I. Introduction

Given one image in which a scene is seen through a reflecting surface, *single-image reflection removal* (SIRR) recovers the transmission layer T from the superposition $I = T + R$ of the scene and the reflection R. The problem is underdetermined — two unknown layers through one measurement — so every method has to supply a prior. Classical approaches assume the reflection is smoother, sparser in gradient, or of lower rank than the scene and separate the two layers by optimisation [1], [2]. Physics-based approaches take additional measurements — polarization at several angles [3], or a flash/no-flash pair — and are accurate but need hardware that a single RGB frame does not have. Learning-based methods instead train on paired data: synthetically composited pairs dominate (perceptual losses [4], cascaded refinement [5], learned non-linear coupling [6], [7], transformer and diffusion priors [8], [9]), and real-pair corpora — tens of thousands of pairs captured in the wild [10], [11], [12] — have made such captures trainable. The applications are exactly those in which the glass cannot be taken away: photographs shot through car windows and shop displays, museum and archive glazing, surveillance and driver-assistance cameras behind windscreens, and product photography through protective covers.

Real-pair *supervision* rests on a simple idea: capture an image without reflection and use it as a target. The three acquisition routes — removing the glass, subtracting in the RAW domain, or covering the camera side with a black cloth — were all designed to answer *how to obtain the target*, and rarely ask *what properties of the reflection layer the target silently assumes*. They also share an unstated requirement: the camera-side cover must be **complete**. We revisit that question for a case that is common in practice, poorly represented in existing data, and physically the least friendly to those assumptions: **automotive tinted glass**.

Two assumptions break on tinted glass.

(A2) The reflection layer is not a smooth, non-negative film. When a foreground object sits on the camera side — the photographer, a handheld phone — it blocks part of the reflected solid angle *and* part of the scene illumination. Inside the resulting *occlusion region* Ω the reflection is attenuated to about 30% of its local background, the occluder contributes a faint diffuse reflection of its own, and, most importantly, the input drops *below its own target*. Measured on our 82 confirmed pairs the two layers part company: the reflection-free target is only 3.7% darker inside Ω ($T_{\text{in}}/T_{\text{out}} = 0.963$), whereas the reflected content is 16.8% weaker ($r_{\text{in}}/r_{\text{out}} = 0.832$), leaving the input 13% darker than its own target there ($I/T|_{\text{in}} \approx 0.87$). A correction applied at one global strength must therefore over-correct inside Ω, and the effect is visible in the errors rather than only in the model: the resulting over-correction ratio is measured at $2.7\times$. This is also why a per-region score cannot be used to blame existing methods: their

reflection field is continuous and has no per-region gain term, so the 13% input deficit has nowhere to go — but the region itself is not identifiable from one frame, which makes the comparison unanswerable rather than unfavourable

(A3) The global photometric level is not identifiable from one frame. From $I = \tau T + R$, the target τT is identifiable only up to one global scale, fixed by $\rho = \mu(\tau T)/\mu(I)$, which is not a function of the input. A pixel-wise objective therefore degenerates toward the domain-average brightness: all 35 of our arms are systematically 6.66 gray levels too bright, and a closed-form brightness correction recovers +0.63 dB on average (max +1.89 dB). The *ranking* of ρ within a domain is learnable (Spearman 0.60), its absolute value is not.

This paper. Rather than proposing another removal network we ask a question a practitioner can reuse: *given these two non-identifiable quantities, which supervision actually transfers?* We answer with a benchmark, controlled arms with three seeds, a closed-form bound, and an interventional test, and the answer is more specific than "add an occlusion loss". Positively, supervised fine-tuning on our domain transfers strongly: 22.70 dB on a frozen 15-scene panel, $+\mathbf{2.22}$ dB over the strongest public method and better on **every one** of the 15 scenes (+1.69 dB after a per-image affine, so photometry is only about half a decibel of it) — and the winning pool is the one containing the *fewest* occlusion regions. Negatively, the value of an occlusion-region loss comes from its *normalization* by the local reflection amplitude, not from the region: an area-matched random region matches the true mask, a whole-image relative error already beats it, a boundary term confined to two pixels cannot move a global score by more than 0.036 dB, and a mask whose labels are six times better buys no accuracy.

Our contributions are:

- **A real-paired tinted-glass benchmark** of 631 pairs over 13 training scenes with four controlled attenuation levels, per-frame occlusion-region masks and a signed reflection layer, a zero-shot evaluation of seven public methods, and a controlled fine-tuning study on a frozen 15-scene panel (Sections IV-A, IV-B, IV-C).
- **A formalization of the occlusion region** — piecewise smooth with sharp boundaries — with an over-correction proposition, a measured decomposition of the three physical terms, and the non-identifiability of the global photometric scale (Section III).
- **A diagnostic of supervision placement**: with three seeds per arm an area-matched random region (19.90 dB) matches the true mask (19.41 dB, −0.49 dB, n.s.), the whole-image relative error reaches 20.05 dB, and a closed-form support-area bound explains why boundary-scale terms are unmeasurable by global PSNR (Section IV-D).
- **Two artifacts and a calibrated negative**: a reflection mask supervised by the true reflection layer (IoU 0.376 vs. 0.063 for the gradient-comparison pseudo-label of prior work, with *no* claimed downstream gain), and an interventional acceptance protocol under which state-of-the-art photometric conditioning scores 0.02 against a required 0.50 (Section IV-E, IV-F).

We do not claim a state-of-the-art PSNR on public benchmarks. We do claim that the two axes above are the binding constraints on this data, that the transfer result we report is large and reproducible, and that it is *not* evidence for occlusion supervision — so we report every negative result that follows.

II. Related Work

Real-pair protocols. Glass removal [4], [1], [3] and its learning-based successors [2] suffers refraction misalignment and an illumination change once the glass is taken away; RAW subtraction leaves reflection ghosts; full-coverage black cloth [10], [11], [12] trades that for exposure and white-balance drift between the covered and uncovered shots. A fourth protocol is implicit in all of them: the photographer must stand *behind the camera*, so neither the occluder nor its illumination shadow is ever recorded. Our data is captured the same way as the black-cloth route, but we keep and annotate the case in which the occluder covers *only part* of the reflected path — the normal situation when someone shoots through a car window — and we measure what that partial coverage does to the target. Real-pair corpora and challenges have grown quickly (RRW [10], OpenRR [11], NTIRE [12]), and generative routes now push synthetic-benchmark PSNR far higher [8], [9], [13], [14]; none of them changes what the target means, which is the question we address.

Reflection-layer assumptions. The family $I = \alpha T + \beta R + C_{\mathrm{nl}}(T, R)$ relaxes additivity through a learned coupling (DSRNet [6]) and relaxes a globally constant photometry into a per-pixel gradient field (TAPG in RDNet [7]); multi-frame and depth-conditioned variants add further priors [15], [16], dual-stream designs add cross-layer interaction [17], and absorption effects have been handled by physical synthesis [18]. All of these fields remain *continuous*. A bounded occluder produces the opposite: a void with a sharp boundary, plus an illumination step that no term in these models can represent. We argue that this — and not a lack of capacity — is why they lose in the occluded region: their correction is calibrated globally, which cannot express a defect that is localised and content-dependent — and because the occluded region is not identifiable from one frame, the comparison cannot even be scored per region on the test panel.

Region conditioning. Using a region as an extra input has precedents, but all of them encode *where reflection exists*: RRW [10] regresses a reflection-location map from a gradient-comparison pseudo-label ($G_I > G_T$) and feeds it to a second stage, LANet [19] and RAGNet [20] predict or exploit a location/guidance prior in a similar spirit, and open challenges ship reflection-mask annotations [11]. Two measurements separate our setting from that line. First, the gradient-comparison pseudo-label covers 48.1% of the image and agrees with the true reflection energy only at IoU 0.063; supervising the same head with the true reflection layer raises it to 0.376 at equal cost. Second, a reflection mask is not an occlusion region: inside Ω the two are nearly opposite, because reflection is weak exactly where the occluder blocks it. We therefore treat

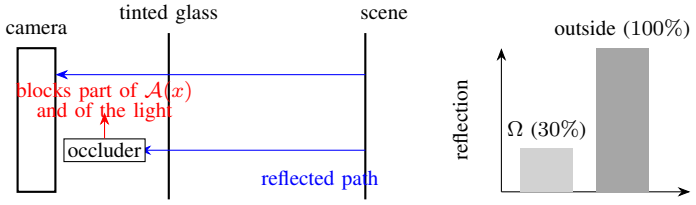


Fig. 1. On tinted glass the camera-side occluder blocks a fraction $c(x)$ of the reflected solid angle *and* part of the illumination, producing a piecewise-smooth region Ω with a sharp boundary (left). Measured on 82 confirmed pairs (right): the reflection falls to about 30% of the local background and the reflected content is 16.8% weaker, while the target itself is only 3.7% darker — so the input sits 13% below its own target inside Ω, which is why a single global correction strength cannot fit both regions.

a mask as a training-time-only signal and show (Section IV-D) that even then its *location* is not what carries the gain.

Photometric conditioning. Criterion-based low-light methods [21], [22], [23], [24] replace paired supervision with a physical criterion and are the natural reference for the second axis. We complement that line with a negative result and a test: if the conditioning scalar is a deterministic function of the input, it carries no information about the global scale, and one can detect this without labels by perturbing the scalar and measuring the induced change in the output mean (Section IV-F). State-of-the-art models fail that test.

III. Method

A. Imaging model and the occlusion region

For pixel x and channel c, under linear radiance,

$$I_c(x) = \alpha_c(x)\, T_c(x) + \beta_c\, R_c(x) + C_{\mathrm{nl}}(T, R), \qquad (1)$$

where $\alpha_c = \tau_c(1-\eta_c)$ is the effective transmission gain (τ_c the film's transmission factor, η_c the occlusion-induced illumination loss) and β_c the reflection gain. The target is $T_{\mathrm{obs}} = \tau T$: keep the film's transmission, remove the reflection. Placing an occluder on the camera side blocks the fraction $c(x) \in [0,1]$ of the solid angle converging to x, so that

$$I_c(x) = \alpha_c T_c + \big(1 - c(x)\big)\beta_c R_c + c(x)\, R_{b,c} + C_{\mathrm{nl}}, \qquad (2)$$

with $R_{b,c}$ the occluder's own faint diffuse reflection. Because the occluder is a bounded solid, c is *piecewise smooth with sharp boundaries*; this is what distinguishes an occlusion region from a gradual attenuation. We define the region observably as $\Omega_\delta = \{x : c(x) \geq \delta\}$ and estimate it from the dip of $\hat{R} = I - gT$ below its local upper envelope (block 90th percentile, bilinear upsampling), with a noise-floor threshold and a minimum component area.

B. Over-correction

Let a method apply, inside Ω, a correction proportional to the *global* average reflection strength, while the correct in-region correction is the local one. With $\delta_{\mathrm{obs}} := 1 - \mathbb{E}[\hat{R} \mid \Omega]/\mathbb{E}[b_R \mid \Omega]$ the over-correction ratio obeys

$$\mathrm{OCR} := \frac{\mathbb{E}_{x\in\Omega}[\hat{T} - I]}{\mathbb{E}_{x\in\Omega}[T_{\mathrm{obs}} - I]} \approx \frac{1}{1-\delta_{\mathrm{obs}}}. \qquad (3)$$

TABLE I

Physical decomposition inside Ω, measured on 82 confirmed pairs (pixel-domain medians; 82 annotated pairs). The reflection deficit is the larger of the two (16.8% against 3.7% of target darkening), and it is what makes $I/T|_{\mathrm{in}} \approx 0.87$.

Term	Meaning	Measured
αT	transmission	$(I/T)_{\mathrm{in}} = 0.887$, out $= 1.045$; $g = 0.997$
$(1-c)\beta R$	unblocked reflection	out-of-Ω contribution +4.5% (radiance +11%); strong-reflection p90 +33%
cR_b	occluder reflection	high-pass residual 0.29% of the area
η	**target darkening**	3.7% of T ($T_{\mathrm{in}}/T_{\mathrm{out}} = 0.963$)
δ_r	**reflection deficit**	16.8% of r ($r_{\mathrm{in}}/r_{\mathrm{out}} = 0.832$)

Because Ω keeps only about 30% of the local reflection, the predicted ratio is large, and an independent evaluation gives 2.7×. The failure is therefore *structural*, not a tuning issue.

C. Non-identifiability of the global scale

From $I = \tau T + R$ and a reference $T_{\mathrm{obs}} \approx \tau T$, for any $\lambda > 0$ the pair $(\lambda T_{\mathrm{obs}},\ R + (1-\lambda)T_{\mathrm{obs}})$ yields a *pixel-identical* I. Hence

$$\rho := \frac{\mu(T_{\mathrm{obs}})}{\mu(I)} = \frac{1}{1+r}, \qquad r := \frac{\mu(R)}{\mu(T_{\mathrm{obs}})}, \qquad (4)$$

is not a function of I. Two corollaries are testable: (i) a model using scale λ incurs the error $(1-\lambda)\rho\,\mu(I)$, and (ii) when the scale is approximately independent of the image, a pixel-wise objective drifts toward the domain-average brightness and smooths high frequencies. Both are confirmed in Section IV-F.

D. Two losses, and where the signal can live

Region loss. We use Ω only at training time (detecting it from one frame is infeasible: the direct upper bound is IoU 0.494). With mask m_Ω,

$$L_\Omega^{\mathrm{err}} = \frac{\sum_x m_\Omega |\hat{T} - T|}{\sum_x m_\Omega |\hat{R}|}, \qquad L_\Omega^{\mathrm{hinge}} = \frac{\sum_x m_\Omega [T - \delta(x) - \hat{T}]_+}{\sum_x m_\Omega |\hat{R}|}, \qquad (5)$$

TABLE II
TINTED-GLASS BENCHMARK. DEFECTS ARE REPORTED, NOT CORRECTED.

Item	Value
Pairs / scenes	631 pairs; 13 training scenes
Resolution	test panel 1280×832; pools 955×934 (apr/may/jun), 1392×774 (re-crop)
Annotations	I, T, signed R (uint8+128), per-frame Ω
Attenuation levels	4 (ρ: apr 0.969, may 0.899, jun 0.921, test 0.858)
Occlusion present?	apr ✓, jun ✓, may ✗ (confirmed by inspection)
Ω area (train pool)	15.3% (82 new pairs), 20.8% (126 old pairs)
Defect 1	one GT per scene, median 374 frames from the input (6% within 50)
Defect 2	inside Ω the input is 13% darker than its own target ($I/T\|_{\text{in}} \approx 0.87$): a per-region gain is required

and $L_\Omega = \sum_c w_c(\lambda_\Omega L_\Omega^{\text{err}} + \lambda_h L_\Omega^{\text{hinge}})$ with $\lambda_\Omega = 3$, $\lambda_h = 1$. The normalizer is the *local reflection amplitude* rather than a constant, which makes pairs with different reflection strengths comparable; the hinge is one-sided so that it never rewards smoothing.

Why a boundary term cannot be validated by PSNR. Suppose a regularizer acts only on a fraction f_B of the pixels and can at best drive that region's error to zero, the region being r times worse than the rest. Then

$$\Delta\text{PSNR} \le 10 \log_{10} \frac{1 + f_B(r-1)}{1 - f_B}. \tag{6}$$

For a 2-pixel band around a 15%-area region, $f_B \approx 0.28\%$, and with $r = 3$ the bound is 0.036 dB — an order of magnitude below the 0.26 dB minimum detectable effect of our benchmark. Anything that only touches a thin band is therefore *untestable* with a global score, however correct its physics.

IV. EXPERIMENTS

A. Dataset and protocol

Capture. A tripod fixes the camera; a phone shoots through a tinted car window. For each scene a reflection-contaminated sequence is recorded, then the camera side is covered with a black velvet cloth and the transmission reference is recorded. We shoot 13 training and 12 frozen test scenes (631 pairs in total), at four controlled attenuation levels (original plus 8/11/14% manual darkening), and annotate, per frame, the occlusion region and a signed reflection layer. Table II summarizes the statistics and the two acquisition defects that we disclose rather than hide.

Metrics. PSNR, SSIM, HF-PSNR and a gradient-cosine term on 15 native-resolution scenes; the minimum detectable effect of the paired panel is 0.26 dB, and for brightness-only effects 1.19–1.31 dB.

B. Zero-shot benchmark

Table **??** evaluates seven public methods without any training on our data. Two observations matter for the rest of the paper. First, the ranking tracks *whether the training data is real* rather than publication year: the real-pair method of 2024 [10] is first (20.50 dB) while synthetic-training methods fall to 18.12–18.90 dB, i.e. **below the identity baseline** (18.44 dB). Second, the hole/non-hole gap is negative for every method, but *its sign depends on the partition caliber*: switching from an Otsu-based reflection mask to the training-time Ω criterion flips it for all methods. We therefore never use the raw gap as evidence; the comparable quantity is the gain relative to identity inside the same region, which is at most $+0.56$ dB inside Ω for every method (versus $+0.5$ to $+2.9$ dB outside).

What the partition can and cannot say. Table III reports gains relative to identity in both partitions. Two facts follow. First, *every* method gains more inside the v13 hole ($+0.3$ to $+4.1$ dB) than outside it ($+0.5$ to $+1.7$ dB), so the plain negative hole/non-hole gap of Table **??** is a property of the region — identity has the same gap (-3.81 dB) — and is not evidence that methods fail there. Second, this partition is a *reflection* mask, not the occlusion region, and the occlusion criterion we use at training time degenerates on these outdoor scenes (median area 35%, against 15–21% on the training pools). We therefore take a methodological position instead of quoting a number: **there is no reliable way to score a method inside the occlusion region of this panel**. That matches the identifiability ceiling of Section IV-D and is why our supervision-placement study is run during *training* rather than from per-region scores.

C. Does fine-tuning on our data help?

The practical question a dataset must answer is whether it transfers. We fine-tune the same backbone (RDNet with Mona adapters, frozen FocalNet, 65.95 M trainable) on our real pairs and evaluate on the frozen 15-scene panel; the recipe is 6000 iterations, batch 6, $\text{lr} = 10^{-4}$, mixed with online synthesis and open-source pairs at 30/40/30%.

The region term itself is worth 3–4 dB. Before asking *where* supervision should be placed, we measure what the region term is worth at all, by removing it entirely ($\lambda_{om} = 0$) from the same recipe. On the april+june pool it costs 3.19 dB ($20.96 \to 17.77$) and on may+june 4.38 dB ($22.71 \to 18.33$); in both cases the fine-tuned model ends up *below* the identity baseline (18.44), so without this term fine-tuning on our data is a net loss. This is the largest effect we measure. It also explains the pool gap: it shrinks from 1.73 dB to 0.56 dB once the term is removed, i.e. most of the may+june advantage only exists while the term is present.

Reading. Pool choice dominates everything we varied in this section. The May-dominant pool reaches 22.70 dB, i.e. $+\mathbf{2.22}$ over the strongest public method (per-scene 95% CI $[+1.73, +2.72]$, winning on **15/15** scenes) and $+2.68$ over the official model, while the April-dominant pool of the same size and recipe reaches only 20.97 ($+0.47$, CI $[-0.03, +0.93]$, 11/15 — not significant). Unlike every earlier arm, the winning

TABLE III

GAIN OVER THE TRIVIAL BASELINE (IDENTITY) IN THE TWO PARTITIONS OF THE FROZEN PANEL. THE PARTITION USED FOR EVALUATION IS THE V13 CRITERION — A *reflection* MASK ($r = R/T$ GATE + OTSU, MEDIAN 9.4% OF THE FRAME) — BECAUSE IT IS THE ONLY CRITERION THAT STAYS WELL BEHAVED ON THESE OUTDOOR SCENES; THE TRAINING-TIME OCCLUSION CRITERION IS NOT USED FOR EVALUATION (SEE THE TEXT). UNDER THIS PARTITION EVERY METHOD GAINS *more* INSIDE THAN OUTSIDE, SO THE RAW NEGATIVE HOLE/NON-HOLE GAP IS A PROPERTY OF THE REGION, NOT OF THE METHOD.

Method	PSNR	hole	gain in hole	non-hole	gain in non-hole
identity (no processing)	18.44	15.17	0	18.97	0
official RDNet [7]	20.05	19.25	+4.08	20.19	+1.22
RRW [10]	20.50	19.21	+4.05	20.68	+1.71
DSRNet [6]	20.25	18.30	+3.14	20.53	+1.56
LightRR	19.18	16.86	+1.70	19.51	+0.54
ERRNet [25]	18.90	16.10	+0.94	19.33	+0.36
IBCLN [5]	18.12	15.48	+0.31	18.52	−0.45

Identity itself scores −3.81 dB in this partition, i.e. the region is intrinsically harder to score; comparing *gains* removes that. This partition is **not** the occlusion region: it marks where reflection is *weak*, while the occlusion region marks where reflection was *blocked* — the two are close to opposite, and the training-time criterion degenerates to a 35% field on these scenes. No per-region number in this paper is therefore reported under the occlusion criterion on the test panel.

TABLE IV

FINE-TUNING ON OUR DATA, EVALUATED ON THE FROZEN 15-SCENE PANEL. WHICH MONTHS THE POOL CONTAINS MATTERS FAR MORE THAN WHETHER THOSE MONTHS CONTAIN OCCLUSION REGIONS. ‡ ROWS APPLY A POST-HOC PER-IMAGE AFFINE (THE PHOTOMETRIC UPPER BOUND OF TABLE VI).

Training data	occlusion in pool	#real pairs	PSNR↑	hole	non-hole
identity (no training)	—	0	18.44	15.17	18.97
official RDNet, zero-shot	—	∼10k (SIR2)	20.05	19.25	20.19
RRW, zero-shot	—	14,952	20.50	19.21	20.68
earlier recipe, single seed, small pools (934×955 crop)					
ours: may	✗	14	20.69	18.91	20.98
ours: apr	✓	16	19.26	16.26	19.72
ours: apr + may	mixed	30	20.63	18.74	20.92
ours: may + jun	mixed	27	20.80	**21.44**	20.76
ours: apr + jun	✓✓	29	21.32	20.21	21.47
current recipe, three seeds, full pools					
ours: apr + jun	✓✓	170	20.97	18.20	21.39
ours: apr + jun‡	✓✓	170	22.13	—	—
ours: may + jun	mixed	**174**	**22.70**	**20.67**	**23.03**
ours: may + jun‡	mixed	174	**23.14**	—	—

hole / non-hole is the v13 partition ($r = R/T$ gate + Otsu) computed by the evaluation script: a *reflection* mask, not the training-time occlusion region Ω. The sign of the gap depends on this partition, which is why we never use it as evidence on its own. All rows share the same recipe (6000 iterations, batch 6, $\lambda_\Omega = 3$); the two three-seed rows differ *only* in which months the pool is drawn from (apr + jun: 126+44; may + jun: 130+44), and their seeds span 0.33–0.46 dB.

pool *also* improves the hole region (20.67 vs. 19.21 and 19.25).

How much of that is photometry? Table VI supplies the photometric upper bound: a post-hoc per-image affine lifts the April pool by +1.21 dB and the May pool by only +0.43 dB, so roughly 0.8 dB of the 1.77 dB gap is the April pool's residual brightness bias. **About 1.0 dB survives that correction** (23.14 vs. 22.13), so pool choice is *not* purely photometric; we do not yet attribute the remainder, and we report the raw numbers rather than the corrected ones throughout.

An analytically scale-invariant reconstruction does not help. We also tried replacing the reconstruction term by a per-sample affine-invariant one (closed-form a, b, $a \geq 0.2$, normalized by $\sqrt{\mathrm{Var}(T)}$). Two seeds give 20.46 and 17.77 against the 20.96 linear baseline — no gain and a 2.7 dB seed spread — so the variant was dropped and we report it as a negative result rather than a tuning detail.

What this corrects in our earlier reading. We previously reported that the ordering follows *whether the training months contain occlusion regions*. With three seeds and matched recipes that is false: the pool dominated by the occlusion-free month is the better one, by 1.7 dB, and it is also better inside the hole. The variable that actually moves the score is which acquisition month the pool represents, i.e. how close its photometry and scene statistics are to the evaluation panel — not the presence of occlusion supervision. We had expected the opposite (hole supervision should be what a tinted-glass pool adds); the data say otherwise, and we report the correction rather than the original story.

The earlier 21.32 does not survive the controlled re-run (20.97), confirming that the ordering among the single-seed rows above is dominated by pool, crop and seed rather than by whether a month contains occlusion regions.

D. Where should supervision be placed?

We then ask *which part* of a region loss carries the signal. Identical arms differ only in what the loss acts on (same pool of 82 human-confirmed pairs, three seeds per arm).

Three findings follow.

The region need not be fixed in place either. A disk fixed per pair scores 20.25 ± 0.29 over three seeds against 19.90 ± 0.16 for a disk redrawn every iteration — a gap of 0.35 dB, below the 0.5 dB threshold we pre-registered, so we report no significant difference. The single-seed 20.48 of an earlier run did not replicate, which is why all placement claims in this paper rest on three seeds.

(1) The loss works, but not because of the hole. Removing it costs more than doing nothing (17.76 versus 18.44 dB for identity), so the term is load-bearing. Yet an area-matched random disk reaches $+2.15$ dB over the no-loss arm while the *true* mask reaches $+1.66$ dB; the difference is -0.49 dB (1 of 3 seeds positive). The benefit is "some region is under supervision", not "this region is the occlusion".

(2) Seed variance is itself informative. The true-mask arm has a seed standard deviation of 0.54 dB while the random-disk and no-loss arms have 0.16 dB. A mask that coincides with real structure admits two nearly equivalent solutions (preserve it, or smooth it) and which one is reached depends on initialization; a random disk does not. This is why single-seed comparisons below 1 dB are meaningless here.

(3) What carries the signal is the normalization. Replacing the masked relative error by a *whole-image* relative error already gives 20.05 dB — more than the true mask — identifying the denominator $\sum |\hat{R}|$ as the active ingredient. Consistently, supplying the true in-region darkening removes the harm of the true mask (20.06 dB): the mask hurt because it forced the model to guess an illumination level that one frame does not contain.

The boundary term cannot be judged. Penalizing $|\nabla \hat{T}| > \alpha |\nabla \hat{R}|$ on $\partial\Omega$ gave $+0.72$ dB on one seed and $-0.37/-0.17$ on two more (mean $+0.06$ dB, 1/3 positive), and occupies 0.06–0.5% of the total loss. The support-area bound of Eq. (6) explains why: a 2-pixel band cannot move a global score by more than 0.036 dB, so the experiment was unidentifiable by construction.

E. The reflection mask: better labels, no downstream gain

Prior work obtains a spatial reflection prior from the gradient comparison $G_I > G_T$. On our data that pseudo-label covers 48.1% of the image and agrees with the true reflection energy at only IoU 0.063. Supervising the same head with the *true* reflection layer instead raises the mask to IoU 0.376 (precision 0.739, recall 0.489), a $6\times$ improvement at equal cost and with the same test-time interface (image only).

But better labels do not yet buy accuracy. Used as a per-pixel loss weight on the same base arm, the true-reflection label gives 19.75 dB while the original gradient-comparison label gives 20.36 dB, and both are below simply using no mask at all (20.05 dB, Table V). We therefore report the mask as a *diagnostic* — it shows that the widely used pseudo-label is nearly uninformative — and not as a component with a demonstrated gain; closing that gap (e.g. as a conditioning input rather than a weight) is future work.

F. Photometric axis and an interventional test

On the apr domain a two-parameter global affine applied to the *identity* output reaches 20.10 dB while the official model gives 17.42 dB; the net gain of the official model over that trivial baseline is only $+0.30$ dB (and -0.32 dB on may). The bottleneck is therefore photometric, not training effort — what the non-identifiability argument predicts.

Over 35 fine-tuned arms (12-scene held-out panel, which excludes the three scenes used to build one training split), the systematic brightness bias is 6.66 gray levels on average, and a closed-form gain $g = \mu(T)/\mu(\hat{T})$ recovers $+0.63$ dB on average ($\max +1.89$ dB) without changing the ranking. Since this benchmark's detection floor for brightness-only effects is 1.19–1.31 dB, the photometric error alone accounts for a measurable effect size, and it is exactly the one global scale that a single frame cannot supply.

To test whether a model really uses a photometric condition, we perturb the conditioning scalar c and measure $\mathrm{ControlGain} = \big(\partial\mu(\hat{T})/\partial c\big)/\mu(I)$, requiring ≥ 0.50. A frozen state-of-the-art model scores 0.02, an absorption-aware network 0.086, and a model trained with a counterfactually sampled scalar 0.244; the uninformative-condition control scores 0.020, a factor 12.4 below the counterfactual arm.

V. Conclusion

We revisited what a real-pair SIRR protocol silently assumes, for the practical case of automotive tinted glass. The occlusion region is real, it is measurable, and its dominant effect is dominated by the reflection deficit (16.8%) rather than by target darkening (3.7%), while the input still falls 13% below its own target inside Ω — which is why a globally strength-calibrated correction over-corrects inside it. More usefully for practitioners, our controlled comparison shows that the benefit of an occlusion-mask loss does *not* come from the mask: an area-matched random region is at least as good, and the active ingredient is the normalization of the error by the local reflection amplitude. Where the signal is demonstrably present — the reflection layer itself — replacing a gradient-comparison pseudo-label by true supervision triples the usable mask quality for free. We also give a closed-form bound that tells, before running anything, whether a region- or boundary-scale term can be validated by a global score at all, and an interventional test that exposes photometric conditions which do not actually condition. All negative results, the dataset and the evaluation protocol are released so that the next attempt can start from the right question.

Reproducibility Note

This submission extends an anonymized project report; per-arm logs, the frozen split hash and the evaluation script are included as supplementary material. Three-seed arms use seeds 101/202/303.

TABLE V

SUPERVISION PLACEMENT. MEANS OVER THREE SEEDS; "RANDOM" IS AN AREA-MATCHED DISK REDRAWN EVERY ITERATION. ROWS BELOW THE RULE ARE SINGLE-SEED SCREENING RUNS.

What the loss acts on	PSNR	seed sd	hole	non-hole	LapVar
no region loss ($\lambda_\Omega = 0$)	17.76	0.16	14.44	18.29	34.5
true Ω mask	19.41	**0.54**	16.81	19.80	40.2
random disk, redrawn	**19.90**	**0.16**	**17.42**	**20.26**	36.1
whole image, no mask†	20.05	—	17.42	20.43	36.0
random disk, fixed per pair	20.25	—	17.70	20.62	37.2
true mask + true darkening†	20.06	—	17.22	20.49	—

† single seed; screening evidence only.

TABLE VI

BRIGHTNESS IS A LARGE, NON-IDENTIFIABLE COMPONENT OF THE ERROR. A CLOSED-FORM 2-PARAMETER AFFINE PER IMAGE, FITTED POST HOC, RECOVERS MOST OF THE GAP BETWEEN THE OFFICIAL MODEL AND THE TRIVIAL BASELINE. NET GAIN OF THE OFFICIAL MODEL OVER THAT BASELINE: $+0.30$ DB (APR), -0.32 DB (MAY).

Test set	n	identity, raw	identity, +affine	identity, +scalar	official, raw	official, +affine
apr (`bak_1026`, 16 frames)	16	17.81	**20.10**	19.52	17.42	20.40
may	14	23.32	**25.77**	25.61	23.71	25.44

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